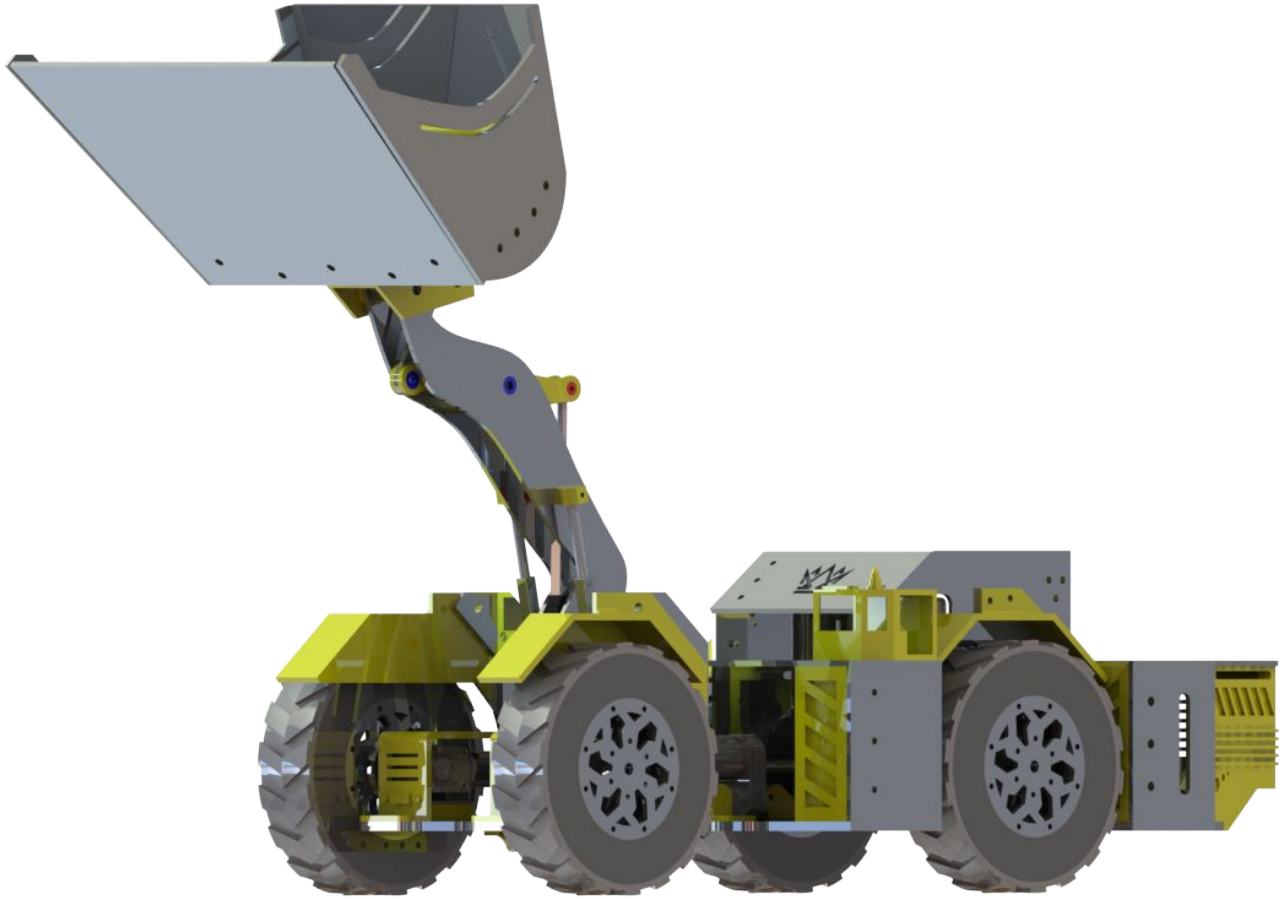
Robotic Load-Haul-Dumper (LHD) scaled system for Mining Automation Technology Development

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Robotic Load-Haul-Dumper (LHD) scaled system for Mining Automation Technology Development

Introduction:

- What is LHD (Load-Haul-Dump machine)?
  - As the name implies, an LHD (Load-Haul-Dump) machine is designed to load mined material, haul it through underground tunnels, and dump it at a designated location.

Manual Load-Haul-Dump (LHD) machines face serious drawbacks:

- High idle time (2–2.5 hrs/shift)
- Operator fatigue and safety risks
- Poor efficiency in constrained environments

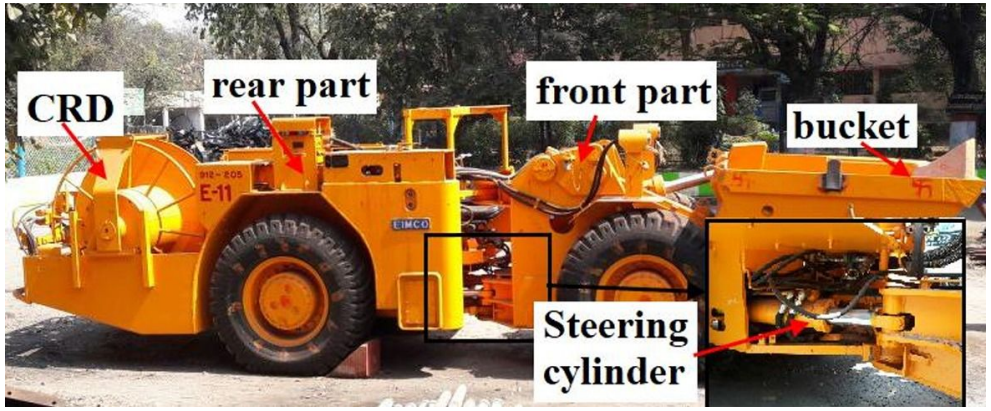


Fig. 1: LHD machine employed at SCCL mine site, AP, India (E912 model, EIMCO ELICON Make)

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Project motivation:

- Surface reserves are depleting — deeper deposits need smarter access.
- There's a growing need for **indigenous** development of automated mining equipment.
- This project proposes a **cost-effective**, **lab-scale**, **smart LHD** designed for "efficiency, safety, and automation".
- Aligns with Industry 4.0 and mining automation goals



Fig. 2: LHD machine of Caterpillar make (R1700 model)

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Objectives:

- Build a **miniature working prototype** of an autonomous LHD machine.
- **Simulate** underground mining operations in a lab setup.
- Incorporate automation using **LiDAR**, **IMU**, and **Jetson Nano**.
- Lay the groundwork for a scalable, Indian-made mining solution.



Fig. 3: LHD machine of Caterpillar make (R3000H model)

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Problem statement:

- India relies heavily on opencast mining — 96% of coal output. [1]
- Underground reserves are underutilized (just 4% contribution). [2]

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Fig. 4: Unsafe environment, not suitable for human operation

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Concept Overview:

The hydraulic circuit of the articulated steering system operation of the LHD machine consists of:

- The main hydraulic pump driving the traction motor, the lift cylinders, steering cylinders.
- The steering valve regulating the desired flow control to the steering cylinders.
- Different sensors to measure the operational state.

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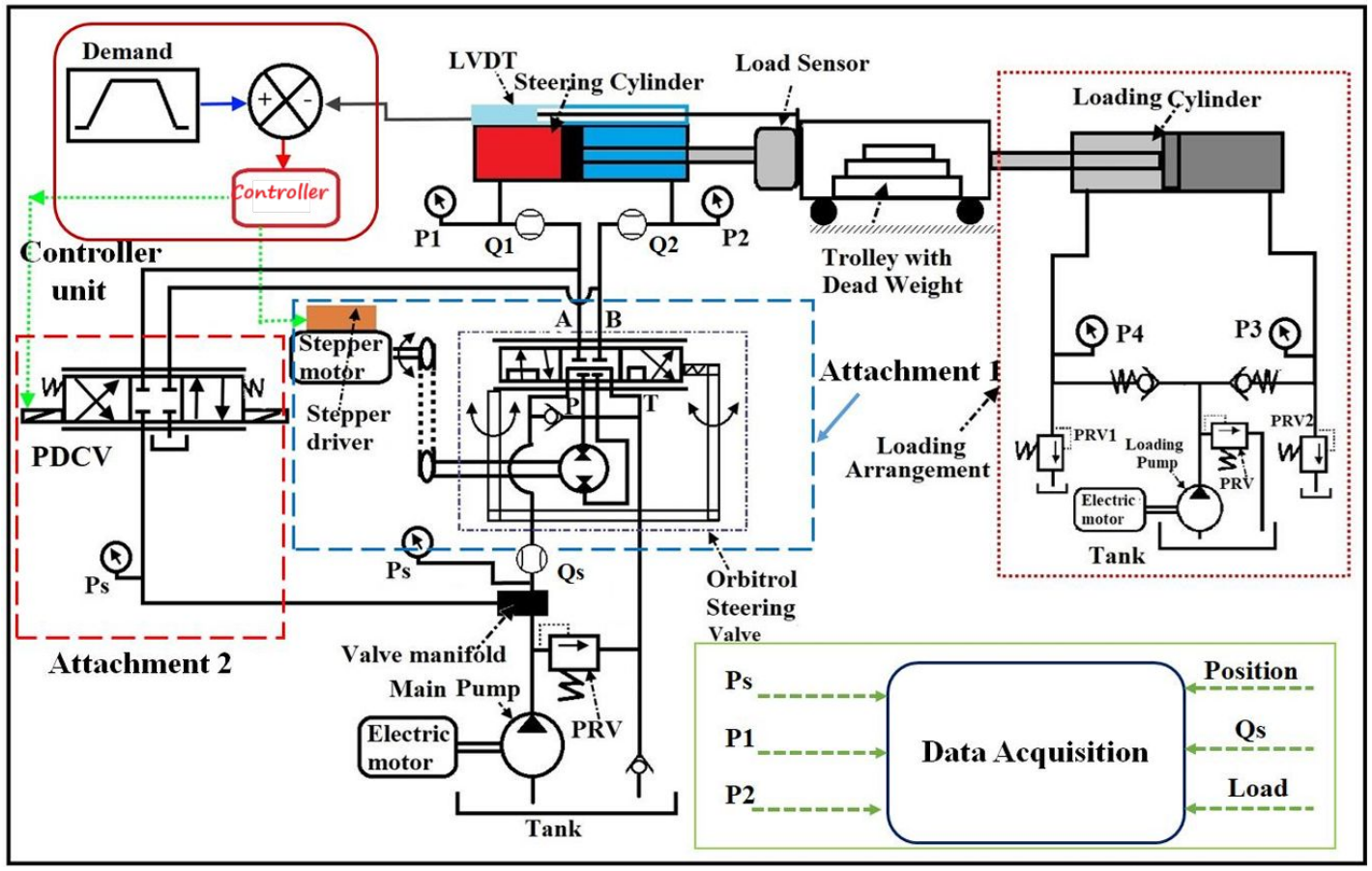


Fig. 5: Simplified hydraulic circuit of the articulated steering actuation through PDCV & Orbitrol valve [3]

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Concept Overview:

Autonomous path planning, and navigation, in static environments has **successfully been demonstrated**:

We propose automation of load haul dumper, via open source software, and newly developed control algorithms, targeted at articulated steering.

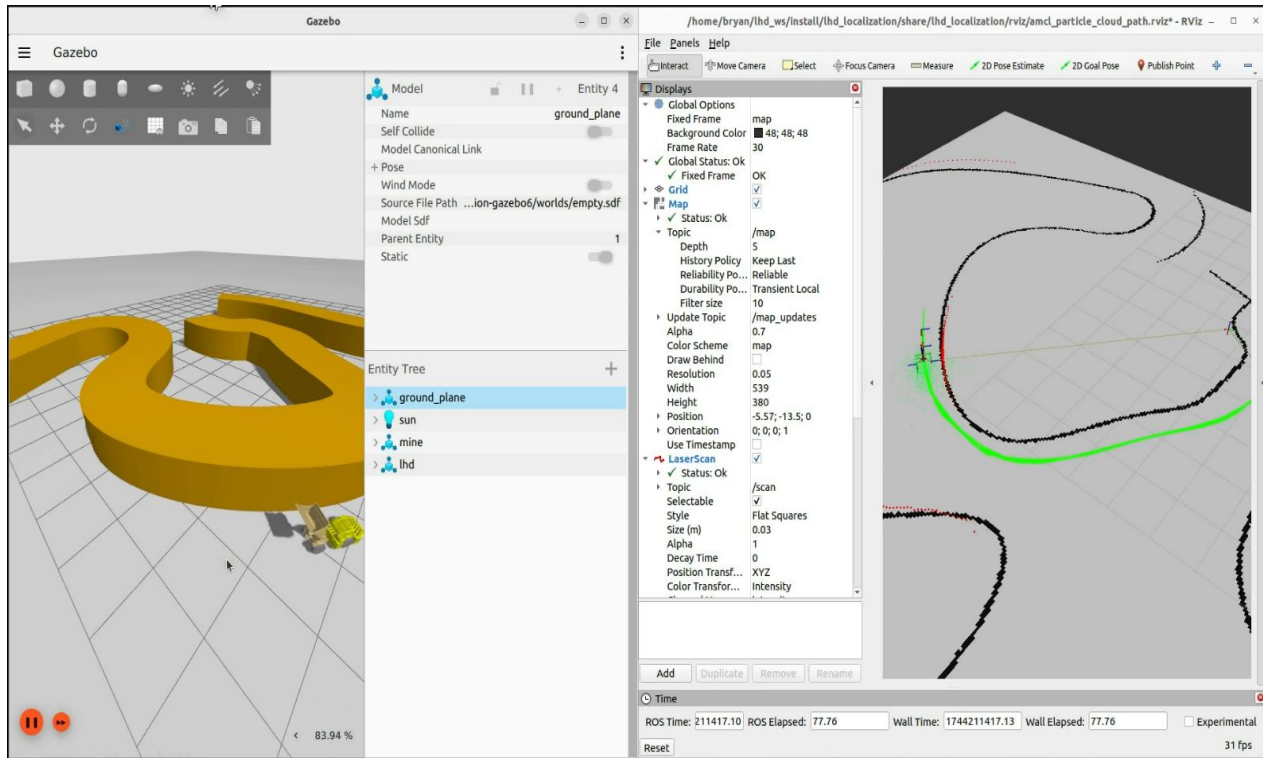


Fig. 6: snapshot of autonomous path traversal in simulation

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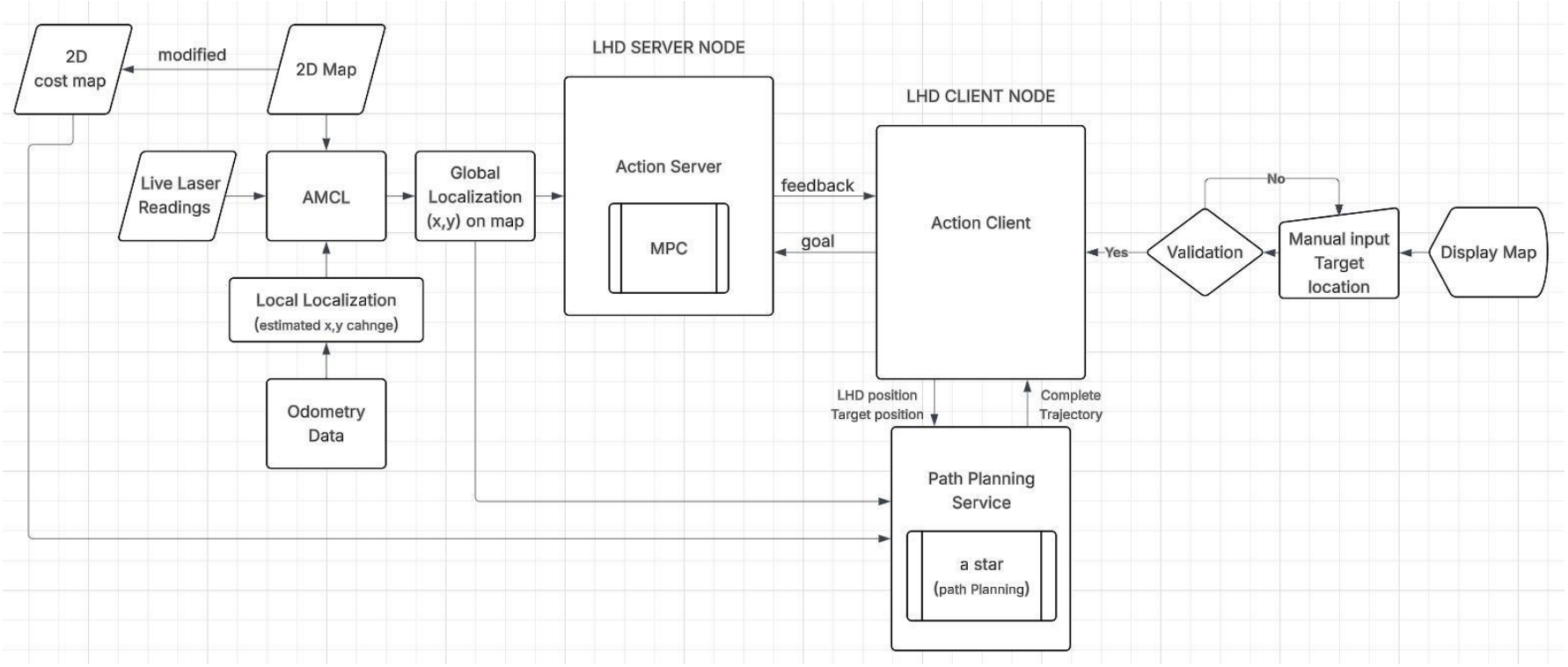
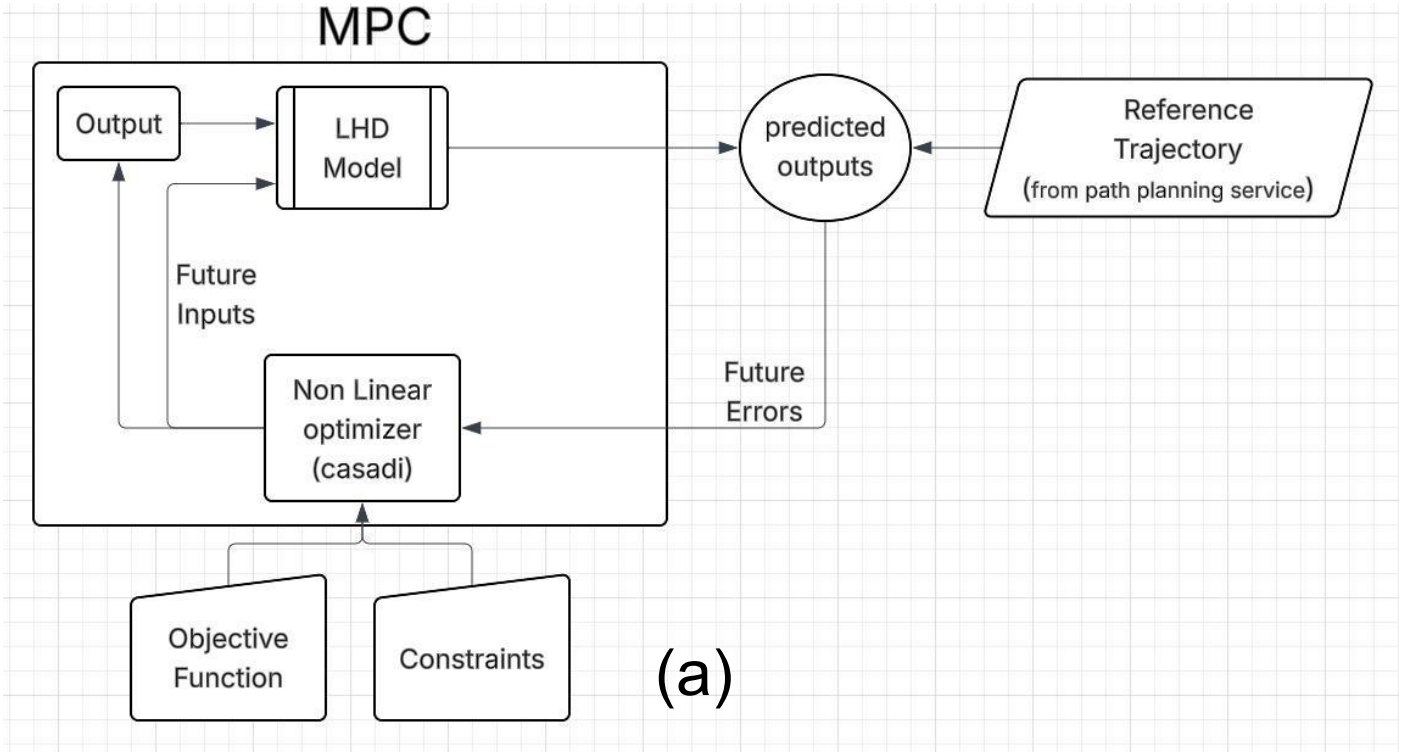
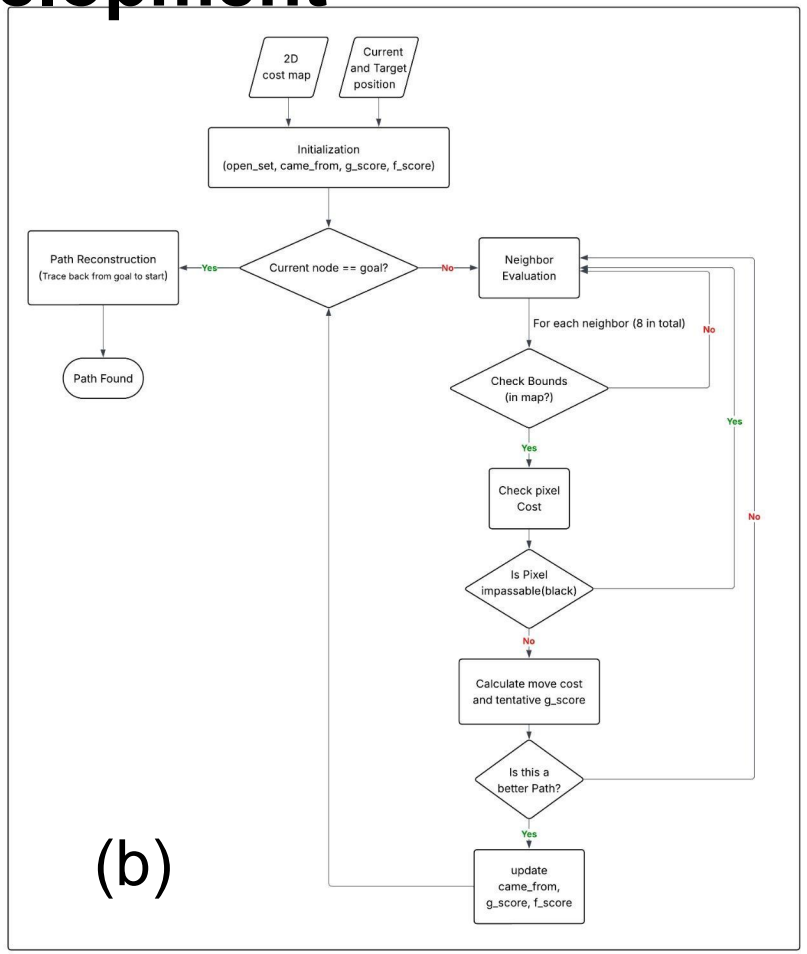


Fig 7: Main Control algorithm Flowchart: (MPC, and a-star blocks on next slide)

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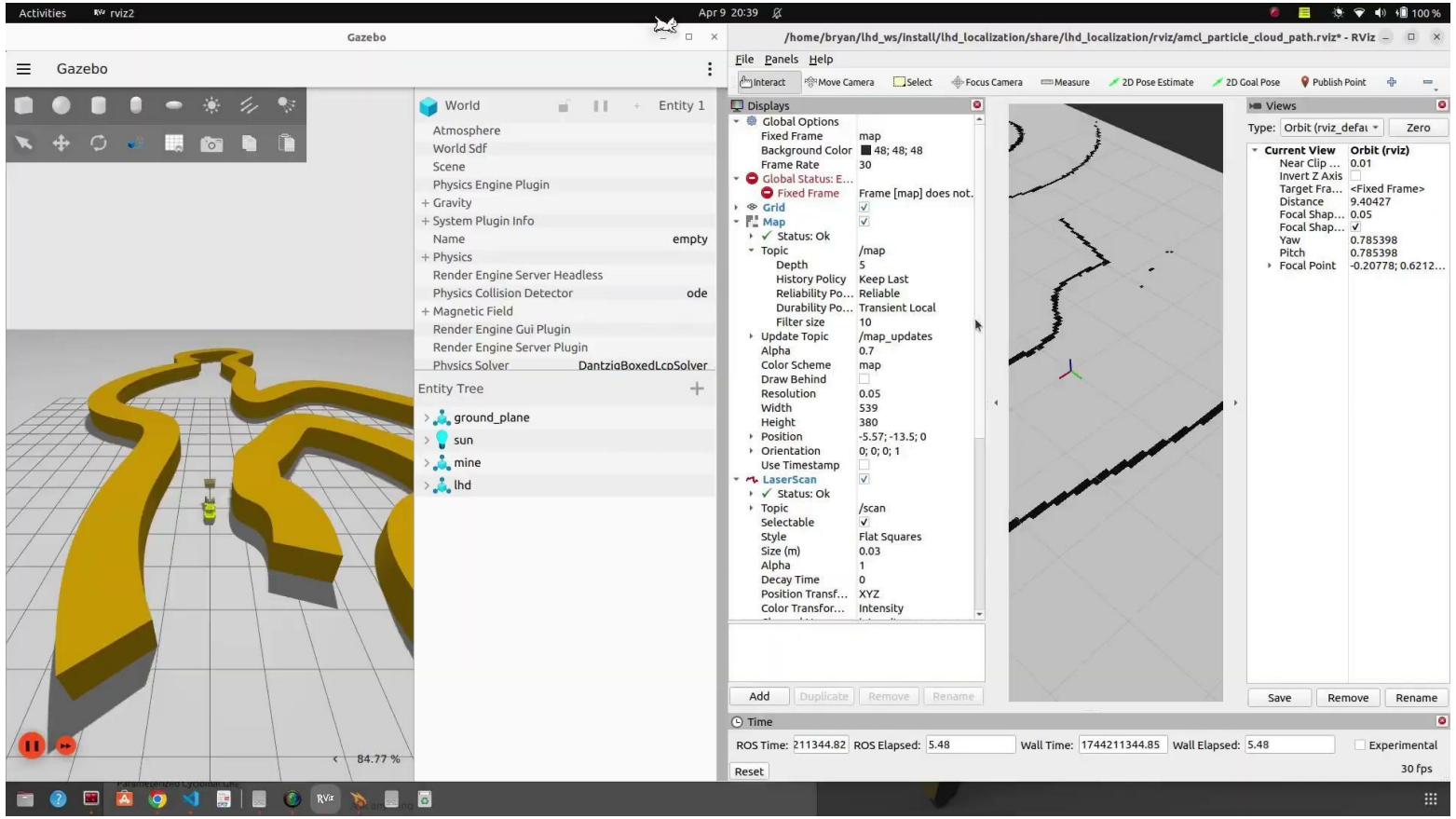
Fig 8. (a) MPC-algorithm Flowchart, (b) a-star





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Demonstration of **Model Predictive Control** based trajectory following, **active Monte-Carlo localisation** and **path planning**, after sending goal location.



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Automation and Control:

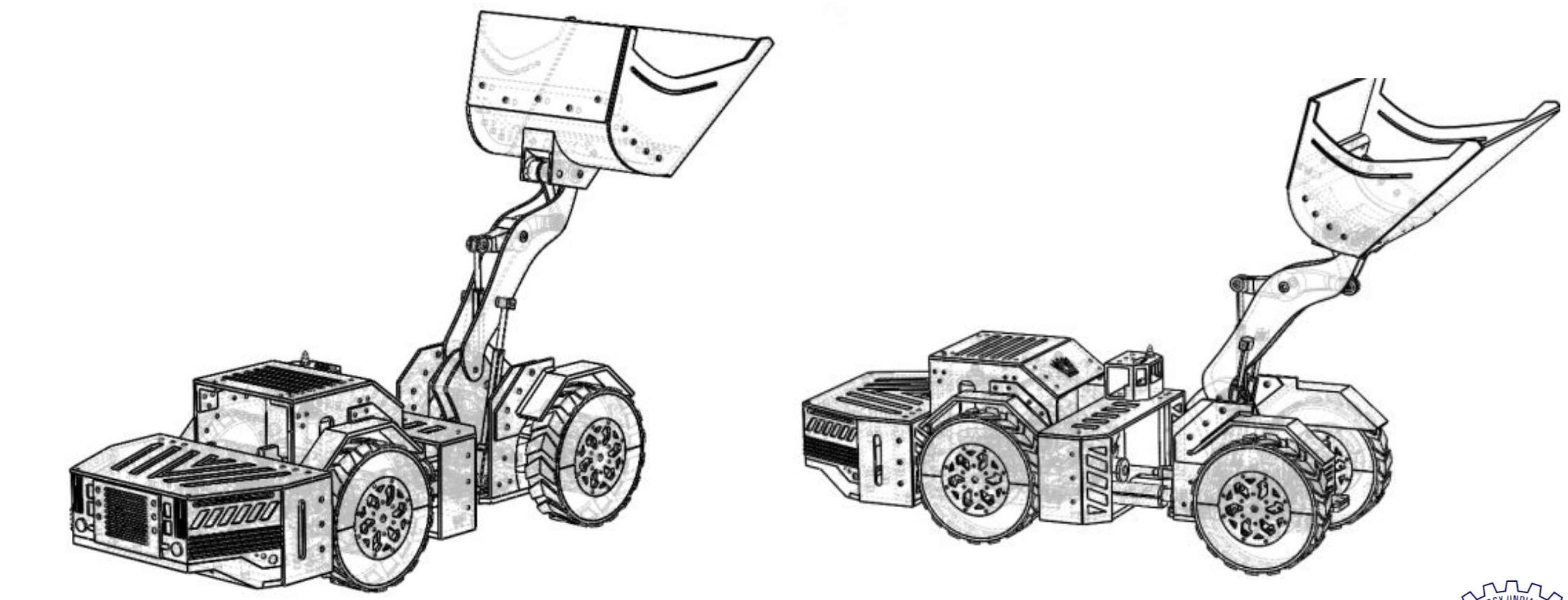
Sensors used: Lidar, Camera, Position sensors for determining actuator length

Controller: Jetson Orion Nano

Navigation strategy: Path creation through Costmap, and A* algorithm, path tracking by MPC, Server for setting target sites in the mine autonomously for LHD to work. (Can also be manual)

Safety features: Emergency stop, dynamic obstacle avoidance.

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Complete model

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Complete model

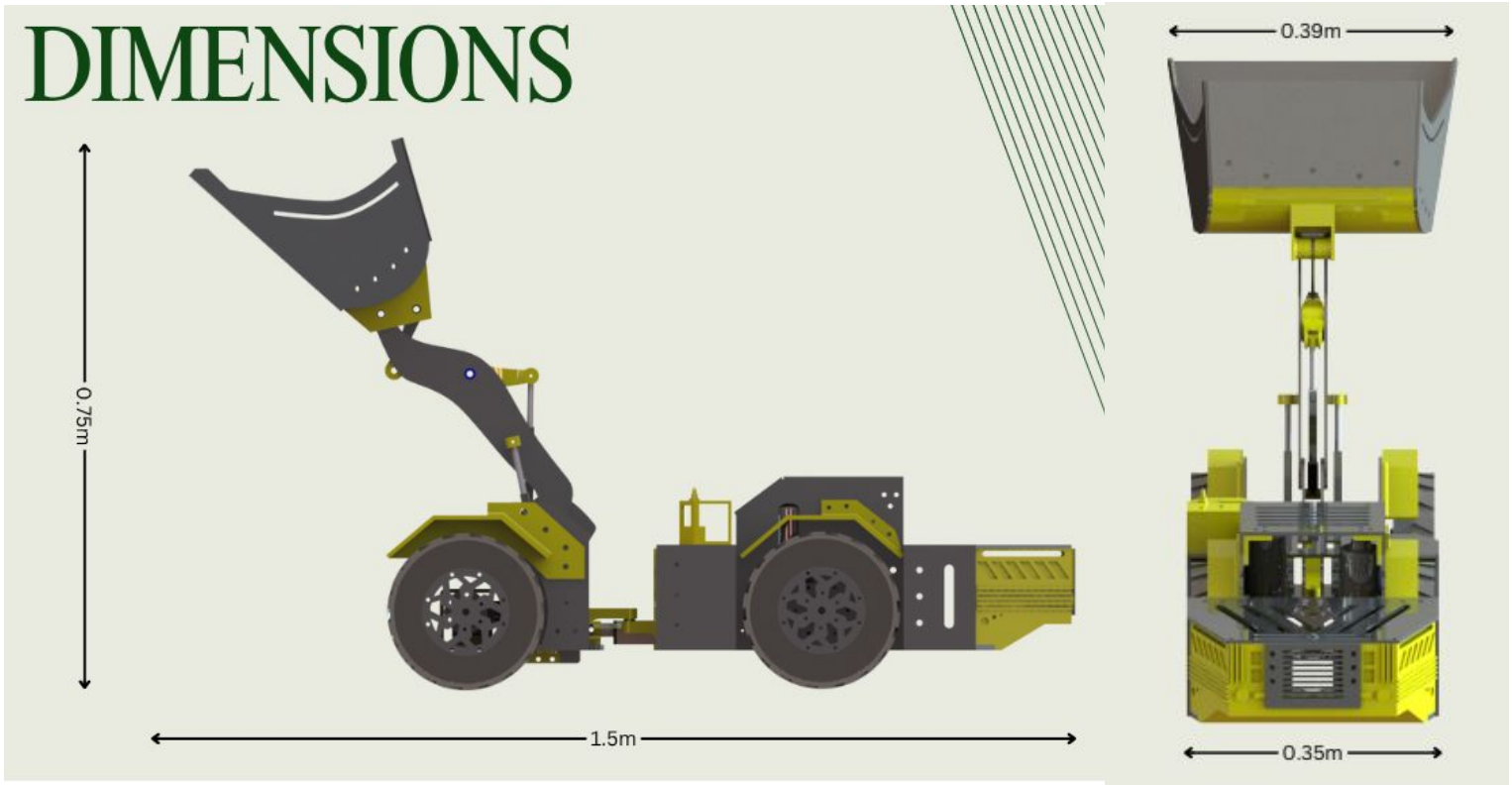


Fig. 9: Dimensions for lab scale prototype

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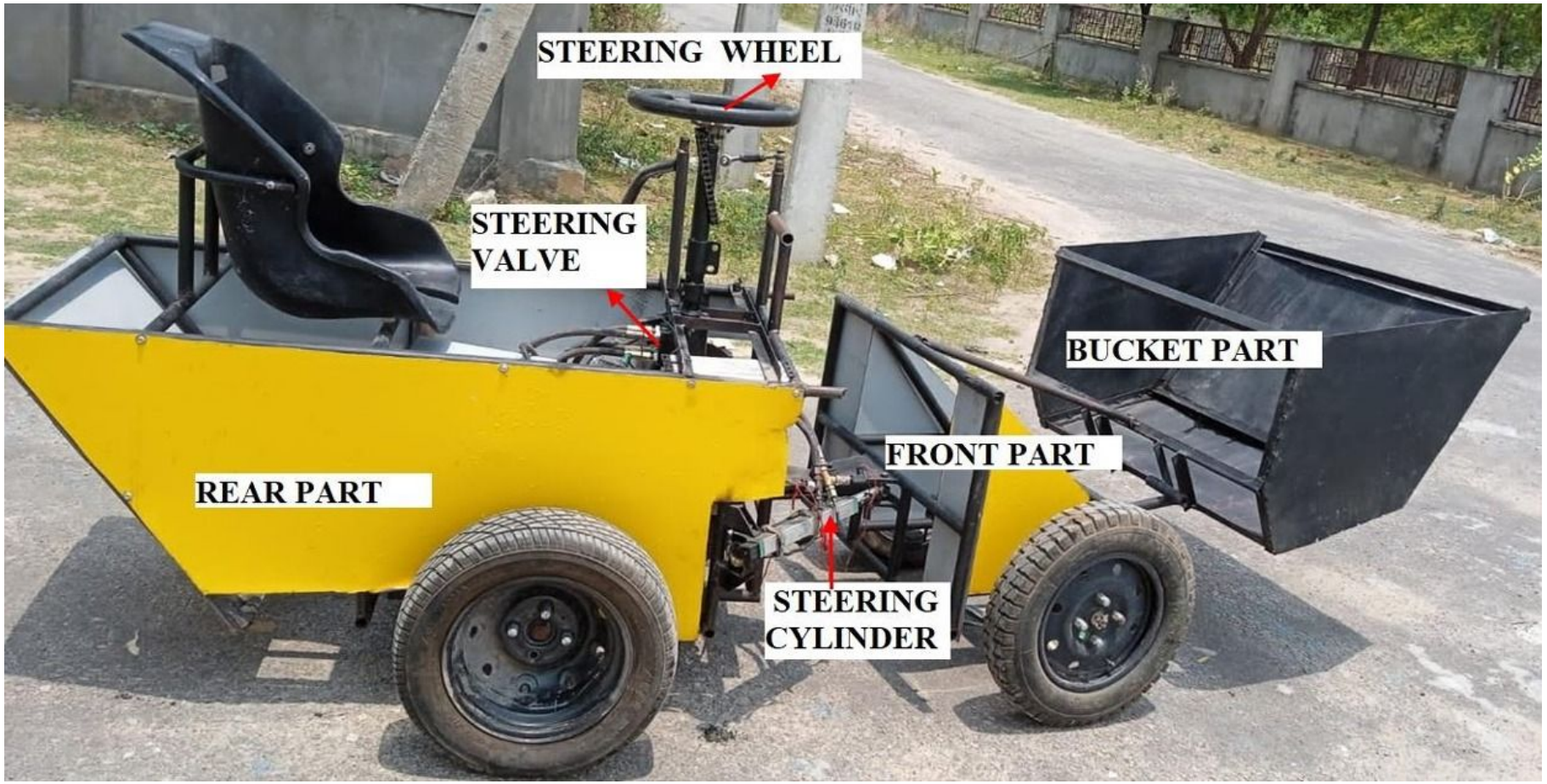


Fig 10: Pictorial view of fabricated lab scale LHD machine

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Innovation and Uniqueness:

- Combines hydraulic efficiency with smart autonomy
- Low-cost and scalable prototype for lab testing
- Enhances skill development in fluid power and automation



Fig 11. Hydraulic Ram

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Potential Impact:

- Safer mining operations with remote-controlled/autonomous LHDs
- Real-time skill training tool in mining institutes
- Adaptable for other underground machines (drills, haul trucks)



Fig 12. Other vehicles may also benefit from this innovation.

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Future works:

- **Development of mine environment** proportional to the lab scale LHD machine.

- Planting the algorithm in the control system to automate the working prototype.

- Further improvements in navigation involve modifying the path planner to account for **dynamic obstacles**[4], and **integration of behaviour trees.**

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Conclusion:

- This project presents a lab-scale Smart LHD that integrates hydraulic efficiency with autonomous control, aiming to revolutionize underground mining.
- It offers a compact, safe, and energy-efficient solution aligned with Industry 4.0 goals, and holds strong potential for future industrial adoption and training applications.

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THANK

YOU